IoT-Based Smart Trolley System Using NodeMCU ESP8266 and RFID for Automated Billing

Project Report

Author: Likhitha C T

Diploma in Electronics and Communication Engineering

GitHub Profile:

https://github.com/LikhithaCT

Technologies Used

- NodeMCU ESP8266
- RFID RC522
- Arduino IDE
- Embedded C
- I2C LCD Display
- Internet of Things (IoT)

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Abstract

- The IoT-Based Smart Trolley System is designed to simplify the shopping experience by automating product identification and billing using RFID technology and the NodeMCU ESP8266 microcontroller.
- Each product is equipped with an RFID tag that is scanned automatically when placed inside the trolley. The system instantly identifies the product, updates the total bill, and displays the information on an I2C LCD screen. Using Wi-Fi connectivity, billing information can also be accessed through a web interface, reducing customer waiting time and minimizing manual billing errors.
- The project demonstrates the practical application of IoT, embedded systems, and wireless communication in retail automation. It offers a cost-effective and scalable solution that improves shopping efficiency while enhancing customer convenience.

- **Keywords:** Internet of Things (IoT), RFID, NodeMCU ESP8266, Smart Shopping, Embedded Systems, Automation.

1. Introduction

- The Internet of Things (IoT) has transformed traditional systems by enabling smart devices to communicate, process data, and automate everyday tasks. In the retail industry, IoT technologies are being adopted to improve customer experience, reduce operational costs, and increase overall efficiency.
- The **IoT-Based Smart Trolley System** is developed to simplify the shopping and billing process by using **NodeMCU ESP8266**, **RFID RC522**, and an **I2C LCD display**. Each product is attached with a unique RFID tag, which is automatically detected when placed inside the trolley. The NodeMCU processes the scanned data, updates the total bill, and displays product details on the LCD in real time.
- The system also provides Wi-Fi connectivity, allowing billing information to be accessed through a web interface. This reduces long billing queues, minimizes manual errors, and improves the overall shopping experience.
- This project demonstrates the practical implementation of embedded systems, RFID technology, and IoT in retail automation. It provides a simple, reliable, and cost-effective solution that can be further enhanced with cloud services, mobile applications, and digital payment integration.

2. Objectives

- The main objectives of the IoT-Based Smart Trolley System are:
- To automate the shopping and billing process using RFID technology.
- To reduce customer waiting time at billing counters.
- To display product details and the total bill in real time.
- To minimize manual billing errors through automatic product identification.
- To enable wireless communication using the NodeMCU ESP8266.
- To provide a user-friendly and efficient shopping experience.
- To demonstrate the practical application of IoT and embedded systems in retail automation.

3. Hardware and Software Requirements

3.1 Hardware Requirements

The Smart Trolley System is developed using the following hardware components:

Component	Description
NodeMCU ESP8266	Wi-Fi-enabled microcontroller used as the main controller.
RFID RC522 Reader	Reads RFID tags attached to products.
RFID Tags	Unique identification tags for products.
I2C LCD Display (16×2)	Displays product details and total bill.
Jumper Wires	Connects all hardware components.
Breadboard	Used for circuit prototyping.
USB Power Supply	Powers the NodeMCU and other modules.

3.2 Software Requirements

The following software tools were used during the development of the project:

Software	Purpose
Arduino IDE	Writing, compiling, and uploading the program.
Embedded C	Programming language used for development.
ESP8266 Board Package	Enables programming of the NodeMCU.
MFRC522 Library	Interfaces the RFID Reader with NodeMCU.
LiquidCrystal I2C Library	Controls the LCD display.

4. System Architecture

The IoT-Based Smart Trolley System consists of four major components: the RFID Reader, NodeMCU ESP8266, LCD Display, and Wi-Fi module.

The RFID reader scans the RFID tag attached to each product when it is placed inside the trolley. The scanned data is sent to the NodeMCU ESP8266, which processes the product information and calculates the total bill. The product details and total amount are displayed on the I2C LCD display. Through the built-in Wi-Fi module, the billing information can also be viewed on a web interface.

Block Diagram

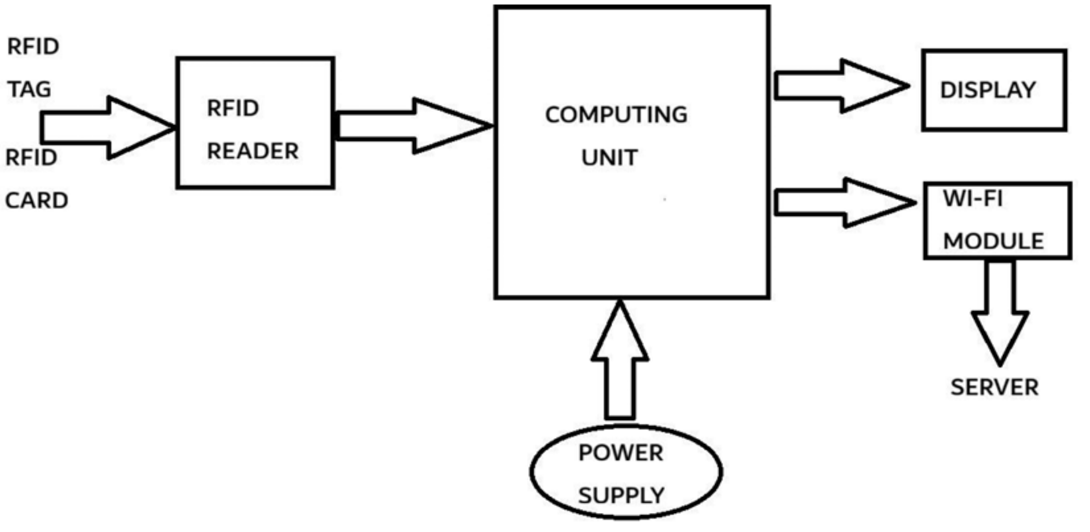

Fig 3.3: Block of IOT Based Smart Trolley System

5. Working Principle

The IoT-Based Smart Trolley System operates by automatically identifying products using RFID technology and updating the billing information in real time.

Step 1: Product Scanning

Each product is attached with a unique RFID tag. When a customer places the product inside the trolley, the RFID RC522 reader scans the tag and retrieves its unique identification number.

Step 2: Data Processing

The scanned RFID data is transmitted to the NodeMCU ESP8266. The microcontroller compares the received tag ID with the stored product database and identifies the corresponding product.

Step 3: Display Information

After successful identification, the product name and price are displayed on the I2C LCD display. The system simultaneously updates the running total of the shopping bill.

Step 4: Wireless Communication

The built-in Wi-Fi module of the NodeMCU enables the billing information to be accessed through a web interface, allowing users to monitor the total bill in real time.

Step 5: Automatic Billing

When shopping is complete, the total amount is already calculated, significantly reducing billing time and eliminating the need for manual scanning at the checkout counter.

6. Results

The IoT-Based Smart Trolley System was successfully designed and tested using the NodeMCU ESP8266, RFID RC522 reader, and I2C LCD display.

The system accurately identified RFID-tagged products, displayed product information on the LCD, and updated the total bill automatically. The Wi-Fi functionality allowed billing information to be viewed through a web interface, demonstrating real-time communication between the trolley and the user.

Project Outcomes

- Successful RFID-based product identification.
- Automatic calculation of the total bill.
- Real-time LCD display of product details.
- Wireless monitoring through Wi-Fi.
- Reduced billing time and manual effort.
- Improved shopping convenience.

7. Advantages

- Reduces customer waiting time at billing counters.
- Eliminates manual billing errors.
- Provides real-time product and billing information.
- Easy to operate and user-friendly.
- Supports wireless communication through Wi-Fi.
- Improves the overall shopping experience.
- Cost-effective solution for retail automation.
- Demonstrates practical implementation of IoT and embedded systems.

8. Applications

The IoT-Based Smart Trolley System can be implemented in various retail environments, including:

- Supermarkets
- Shopping Malls
- Retail Stores
- Grocery Stores
- Department Stores
- Smart Retail Chains
- Automated Shopping Systems

9. Future Enhancements

The current prototype can be further improved by integrating advanced technologies to enhance functionality and user experience.

Future enhancements include:

- Mobile application for shopping and billing.
- QR code and digital payment integration.
- Cloud database for inventory management.
- AI-based product recommendation system.

- Weight sensors for additional product verification.
- Voice assistance for customers.
- Barcode scanner integration.
- Customer loyalty and reward system.

10. Conclusion

The IoT-Based Smart Trolley System demonstrates an effective application of IoT and embedded systems in retail automation. By integrating the NodeMCU ESP8266, RFID RC522 reader, and LCD display, the system successfully automates product identification and billing.

The project minimizes manual effort, reduces billing time, and improves customer convenience through real-time product tracking and wireless communication. It also provides valuable experience in embedded programming, IoT communication, RFID technology, and hardware integration.

Overall, the project offers a simple, reliable, and cost-effective solution for smart retail systems and provides a strong foundation for future enhancements using cloud computing, mobile applications, and artificial intelligence.

References

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